

Week - 05 - Classes, Graphics\Lecture_10_code_4.py

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1  '''
2  Inheritance: Classes in a child parent relationship
3  Class Projectile is a parent (or super or base) class. Missile, Bomb and Rocket classes
   derive from Projectile classes, that means Missile, Bomb and Rocket are children of
   Projectile class.
4  Also see how Projectile, Missile, Bomb and Rocket classes have a method with the same name
   'get_projectile_data'. It is a basic example of polymorphism, that means, multiple
   functions with the same name. Which function is actually called is decided based on
   arguments, which is 'self' in this case
5  '''
6  import math
7  import time
8
9  G = 9.82
10 AIR_DRAG = 0.4
11
12 class Projectile:
13     def __init__(self, ID, size, energy):
14         self.id = ID
15         self.size = size
16         self.energy = energy
17         self.x_pos = 0
18         self.y_pos = 0
19
20     def get_projectile_data(self):
21         # A getter function
22         return (f'id {self.id}, size {self.size}, energy {self.energy} and position
   ({self.x_pos}, {self.y_pos})')
23
24
25 class Missile(Projectile):
26     def __init__(self, ID, size, energy, feul, guidance_system):
27         super().__init__(ID, size, energy)
28         self.feul = feul
29         self.guidance_system = guidance_system
30
31     def get_projectile_data(self):
32         base_data = super().get_projectile_data()
33         return base_data + self.guidance_system
34
35     def guide(self):
36         print("Missile is on a guided path")
37
38 class Bomb(Projectile):
39     def __init__(self, ID, size, energy, explosion_method):
40         super().__init__(ID, size, energy)
41         self.explosion_method = explosion_method
42
43     def get_projectile_data(self):
44         base_data = super().get_projectile_data()
45         return base_data + self.explosion_method
46
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47     def explode(self):
48         print("Bomb will explode with a timer")
49
50 class Rocket(Projectile):
51     def __init__(self, ID, size, energy, speed):
52         super().__init__(ID, size, energy)
53         self.speed = speed
54
55     def get_projectile_data(self):
56         base_data = super().get_projectile_data()
57         return base_data + str(self.speed )
58
59     def show_speed(self):
60         print("Current speed of rocket is", self.speed)
61
62 m1 = Missile('M101', 25, 23, 'GPS', 'solid')
63 r1 = Rocket('R101', 15, 13, 500)
64 b1 = Bomb('B101', 12, 8, 'Timer')
65
66 print('missile data: ', m1.get_projectile_data())
67 print('rocket data: ', r1.get_projectile_data())
68 print('bomb data: ', b1.get_projectile_data())
69
70 b1.explode()
71 m1.guide()
72 r1.show_speed()
73
```